PES University, Department of Computer Science and Engineering

# SE Lab 1: Requirements Engineering and UML Use-Case Modelling

## Problem Statement 44: Database Query Performance Profiler

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| Req ID | Type | Description | Priority | Acceptance Criteria | Rationale (short) | Comments |
|---|---|---|---|---|---|---|
| **FR-001** | Functional | The system shall parse EXPLAIN plans from Postgres and MySQL, mark every sequential scan on a table that is above the configured row limit, and suggest an index for it (which columns, in what order, and which index type). | High | Pass: A query that filters on two columns of a table of about one million rows with no index on them is picked up, and the tool suggests a composite index on both of those columns.<br>Fail: A plan that still shows a sequential scan on a large table is reported as fine. | This is the main job of the tool. Without it we only learn that a query is slow, not what to do about it. | |
| **FR-002** | Functional | The system shall collect slow query logs from every registered database, replace the literal values in each statement with placeholders so that similar queries form one fingerprint, and store the call count, total time and average time for each fingerprint. | High | Pass: A log file of 1,000 lines containing 50 different statement shapes is grouped into exactly 50 fingerprints, and the call counts add up to the counts in the log file.<br>Fail: The same query written with different values is stored as two separate fingerprints. | Raw logs are too noisy to rank. Similar queries have to be grouped before anything useful can be said about them. | |
| **FR-003** | Functional | The system shall let a Database Administrator add a database and set its profiling options, namely the slow query threshold in milliseconds, how many days logs are kept, and the row count above which a table is treated as large. | Medium | Pass: After the threshold is saved as 500 ms, the next collection cycle picks up a query averaging 501 ms and ignores one averaging 499 ms, without restarting the service.<br>Fail: The new threshold is applied only after the service is restarted. | Every database has a different idea of what counts as slow, so this cannot be fixed in the code. | |
| **FR-004** | Functional | The system shall build a weekly digest listing the ten fingerprints with the highest total execution time, and for each one show the current plan cost, the suggested index and the expected saving, and then send the digest to the Backend Lead. | High | Pass: The digest is generated every Monday morning, has at most ten entries sorted by total execution time, shows an expected saving for each entry, and reaches the Backend Lead within five minutes.<br>Fail: The digest is sent with entries in no particular order, or with the suggestion column left empty. | Nobody watches a dashboard all week. A short weekly summary is the thing that actually gets read and acted on. | |
| **FR-005** | Functional | The system shall let a user mark any suggestion as Accepted, Rejected or Deferred along with a short comment, and shall keep rejected suggestions out of later digests for a period chosen by the user. | Medium | Pass: A suggestion rejected for 30 days does not appear in the next four weekly digests, and the record stores who rejected it and when.<br>Fail: A rejected suggestion shows up again in the very next digest. | If the same rejected index keeps coming back every week, people stop reading the digest altogether. | |

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|---|---|---|---|---|---|---|
| **NFR-001** | Nonfunc. (Performance and Security) | The system shall handle up to 5,000 slow query log records per minute without losing any record and without pushing the CPU usage of the monitored database up by more than 5 percent. | High | Pass: In a 30 minute run at a steady 5,000 records per minute, no record is lost, the backlog stays under 10 seconds, and the extra CPU load on the host database stays within 5 percent.<br>Fail: The backlog keeps growing through the run, or any log record goes missing. | A profiler that slows down the database it is supposed to be watching is worse than having no profiler at all. | |
| **NFR-002** | Nonfunc. (Security) | The system shall store database passwords encrypted with AES-256, replace literal values in stored queries with placeholders, and allow only the DBA role to change settings or run a simulation on the replica. | High | Pass: A dump of the tool's own storage shows no readable password, a sample of 200 stored queries shows placeholders instead of real values, and a Backend Lead account is refused with a 403 on the settings pages.<br>Fail: Any readable password, or any real customer value, is found in storage or in the log files. | Slow query logs carry real production data, and the tool itself holds a login to the database, so it is a tempting target. | |

**Actors:** Database Administrator and Backend Lead are the two people who use the tool. The Monitored Database Server, the Report Scheduler and the Notification Service are supporting actors that the system talks to.

**Use cases taken from these requirements:** UC-01 Ingest Slow Query Logs (FR-002), UC-02 Analyse SQL Execution Plan (FR-001), UC-03 Recommend Index Candidates (FR-001, FR-005), UC-04 Generate Weekly Digest (FR-004), UC-05 Configure Profiling Settings (FR-003), UC-06 Fingerprint and Deduplicate Query, UC-07 Simulate Index on Replica, UC-08 Export Digest as PDF/CSV.

*The Comments column has been left empty for the peer review step in Lab 1.*